2. (a) It means Q is square ($n \times n$), nonsingular, and its columns form an orthonormal basis of $\mathbb{R}^n$. As a byproduct of all this, $Q^T = Q^{-1}$, and Q^T is also orthogonal.

(b) $$\|Q\vec{x}\|^2 = \langle Q\vec{x} \mid Q\vec{x} \rangle = (Q\vec{x})^T Q\vec{x} = \vec{x}^T Q^T Q \vec{x}$$

$$= \vec{x}^T I \vec{x} = \langle \vec{x} \mid \vec{x} \rangle = \|\vec{x}\|^2.$$

Thus, $\|Q\vec{x}\| = \|\vec{x}\|$.

3. (a) $$\det(A) = -1 \begin{vmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \\ 2 & -1 & 3 \end{vmatrix} = (-1)(0) = 0$$

(b) $$\det(B) = \begin{vmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \\ 2 & -1 & 3 \end{vmatrix} + \begin{vmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & 3 \end{vmatrix} = -\det(A) + (2)(4)(3) = 24$$

(c) $$\det(C) = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) \begin{vmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \\ 2 & 1 & 6 \end{vmatrix} = \frac{1}{6}\det(B) = 4.$$

(d) $$\det(BC^{-1}) = \frac{\det(B)}{\det(C)} = \frac{24}{4} = 6$$

4. (a) Take $$\vec{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$ Then take $$\vec{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{2}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Finally, take

$$\tilde{u}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - 0\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \\ -1/2 \\ 0 \end{bmatrix} \quad \left(\times 2 \to \vec{u}_3 \right)$$

Thus $$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}$$ is an orthogonal basis for W.

(b) Call this closest vector w. Then

$$w = \text{proj}_W \vec{v} = \text{proj}_{\vec{u}_1} \vec{v} + \text{proj}_{\vec{u}_2} \vec{v} + \text{proj}_{\vec{u}_3} \vec{v}$$

$$= \frac{2}{2}\vec{u}_1 + \frac{5}{1}\vec{u}_2 + \frac{12}{6}\vec{u}_3$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + 5\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

5. (a) Linear independence implies there can be no free columns in A, and (at a minimum) this means there are at least as many rows in A as there are columns. Thus, $m \geq n$.

(b) The mapping $\vec{x} \mapsto A\vec{x}$ takes an $\vec{x}$ from $\mathbb{R}^n$ to the column space of A, a subspace of $\mathbb{R}^m$. This particular $\vec{x}_0 = (A^TA)^{-1}A^T\vec{b}$ is the preimage of the vector in $\text{Col}(A)$ closest to $\vec{b}$. Because A has linearly independent columns, $\vec{x}_0$ is the only vector in $\mathbb{R}^n$ with this property.

6.

$$p(A) = p(V\Lambda V^{-1}) = (V\Lambda V^{-1} - \lambda_1 I)(V\Lambda V^{-1} - \lambda_2 I)\cdots(V\Lambda V^{-1} - \lambda_n I)$$

$$= V(\Lambda - \lambda_1 I)V^{-1}V(\Lambda - \lambda_2 I)V^{-1} \cdots V(\Lambda - \lambda_n I)V^{-1}$$

$$= V\begin{bmatrix} 0 & & & 0 \\ & \lambda_2-\lambda_1 & & \\ & & \lambda_3-\lambda_1 & \\ 0 & & & \ddots \; \lambda_n-\lambda_1 \end{bmatrix}\begin{bmatrix} \lambda_1-\lambda_2 & & & 0 \\ & 0 & & \\ & & \lambda_3-\lambda_2 & \\ 0 & & & \ddots \; \lambda_n-\lambda_2 \end{bmatrix}\cdots\begin{bmatrix} \lambda_1-\lambda_n & & & 0 \\ & \lambda_2-\lambda_n & & \\ & & \lambda_3-\lambda_n & \\ 0 & & & \ddots \; 0 \end{bmatrix}V^{-1}$$

$$= V\begin{bmatrix} 0 & & & 0 \\ & 0 & & \\ & & \ddots & \\ 0 & & & 0 \end{bmatrix}V^{-1} = 0.$$

7. Let $\vec{u}_1, \vec{u}_2, \vec{u}_3$ be the vectors $\begin{bmatrix} 1 \\ -2 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 9 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ 13 \end{bmatrix}$, respectively. Because they are a basis of $\mathbb{R}^3$, given any $\vec{x} \in \mathbb{R}^3$ there exist scalars $\vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ such that

$$\vec{x} = c_1\vec{u}_1 + c_2\vec{u}_2 + c_3\vec{u}_3 = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix}\vec{c},$$

and for the matrix A of the transformation,

$$A\vec{x} = A(c_1\vec{u}_1 + c_2\vec{u}_2 + c_3\vec{u}_3) = c_1 A\vec{u}_1 + c_2 A\vec{u}_2 + c_3 A\vec{u}_3$$

$$= \begin{bmatrix} A\vec{u}_1 & A\vec{u}_2 & Au_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 6 \\ -1 & 1 & 5 \\ -2 & 1 & 1 \end{bmatrix} \vec{c}.$$

But $\vec{c} = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix}^{-1} \vec{x}$, so $A\vec{x} = \begin{bmatrix} 1 & 3 & 6 \\ -1 & 1 & 5 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -2 & -3 & -4 \\ 6 & 9 & 13 \end{bmatrix}^{-1} \vec{x}$,

Thus, $$A = \begin{bmatrix} 1 & 3 & 6 \\ -1 & 1 & 5 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -2 & -3 & -4 \\ 6 & 9 & 13 \end{bmatrix}^{-1}.$$